

Continuous-Time Stabilization by Compact Memory: A Counterexample Suggested by arXiv:2211.15121

Automatic mathematics run `fa_banach_001`

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Status. Candidate counterexample, likely valid. At the continuity level of the source’s discrete example, a positive essential spectral bound, a compact nonlinear perturbation, and a quadratically small perturbation do *not* force continuous-time instability. The compact map below is not locally Lipschitz at zero, so stronger regularity variants are not addressed.

1 Source question

The source is V. Müller, R. Schnaubelt, and Y. Tomilov, *On growth and instability for semilinear evolution equations: an abstract approach*, arXiv:2211.15121v3 [1]. Its Example 5.3 constructs a stable discrete system

$$x_{n+1} = (A + K + G)x_n$$

on an infinite-dimensional Hilbert space, although $r_e(A) > 1$, K is a continuous compact map, and $G(x) = O(\|x\|^2)$ at zero. Immediately afterward, on printed page 19, the authors write:

“We are short of a similar construction for continuous time. However, we suspect that an example similar to the above for (1.1) may not exist.”

The same page recalls the continuous equation and the paper’s definition of stability. The complete context is reproduced below.

example for $\varepsilon = \varepsilon_k$. Using the estimate (3.4), we infer that $\|y_{n_0}\| \geq \frac{\varepsilon_k}{2}$, a contradiction. We have shown that

$$\sup_n \|y_n\| \leq \frac{\varepsilon_k}{2} \quad \text{and} \quad \sup_{j,n} f_j(\|y_n\|) \leq \frac{\varepsilon_k}{2},$$

so that $\sup_n \|x_n\| \leq \varepsilon_k$. Now to prove the stability of the zero solution to (5.1), given $\epsilon > 0$, it suffices to consider $\epsilon_k \in (0, \epsilon)$, and to choose $\delta = \epsilon_k^3/4$.

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6. WELL-POSEDNESS FOR CONTINUOUS TIME AND GLOBAL EXISTENCE

We are primarily interested in nonlinear evolution equations of the form

$$(6.1) \quad x'(t) = Ax(t) + K(t, x(t)), \quad x(0) = x_0, \quad t \in [0, T],$$

where $T \in (0, \infty]$, A generates a C_0 -semigroup $(T(t))_{t \geq 0}$ on a Banach space X , and $K : [0, \infty) \times X \rightarrow X$ is a function such that $x \mapsto K(t, x)$ is compact for fixed t and that satisfies some additional regularity assumptions. One can study classical solutions of (6.1), that is, maps $x \in C^1([0, T], X) \cap C([0, T], D(A))$ solving (6.1). However, it is often convenient to deal with the integrated version of (6.1), namely

$$(6.2) \quad x(t) = T(t)x_0 + \int_0^t T(t-s)K(s, x(s)) ds, \quad t \geq 0.$$

Recall that a continuous function $x : [0, T] \rightarrow X$ is called a *mild* solution of (6.1) if it satisfies (6.2). Each classical solution of (6.1) is a mild solution, but not conversely. A mild solution is a classical one if $x_0 \in D(A)$ and, for instance, $K \in C^1([0, T] \times X, X)$. In the sequel we only treat mild solutions, which we will call *solutions* from now on, for simplicity.

If X has nice geometric properties, e.g., if it is reflexive, then mild solutions can be identified with so-called weak solutions of (6.1), that is, $x \in L^1([0, T], X)$ satisfying (6.1) in a weak sense. This notion will not be used in this paper, however.

Let the solution $x(t) = x(t, x_0)$ of (6.2) exist for all $t \geq 0$, i.e., one has $T = \infty$. It is called (nonlinearly) stable if for every $\varepsilon > 0$ there is radius $\delta > 0$ such for all $y_0 \in B(x_0, \delta)$ all solutions y with $y(0) = y_0$ are defined on $[0, \infty)$ and satisfy $\|y(t) - x(t)\| < \varepsilon$ for all $t \geq 0$. Without loss of generality, one may assume here that $x_0 = 0$. If the solution x is not stable, then it is said to be unstable.

Instability is a local and rather weak property. It is often of interest to show *global* properties of x which are stronger than mere instability. Such properties are the main topic of this paper. To not overshadow the study of

The precise analogue tested here is whether the zero solution of

$$x'(t) = Ax(t) + K(x(t)) + G(x(t)) \tag{1}$$

must be unstable when the essential spectral bound of A is positive, K is continuous and compact with $K(0) = 0$, and $G(x) = O(\|x\|^{1+\delta})$ for some $\delta > 0$. The answer is no.

2 Idea of the proof

The compact perturbation uses only one scalar “memory” coordinate. If y is the component in the infinite-dimensional unstable subspace, the memory grows at speed $\sqrt{\|y\|}$. It therefore becomes much larger than $\|y\|$ near zero. A damping term of size at most $2|a|^2$, hence quadratic in the full state, can then reverse the growth of y before it leaves a small neighborhood.

After setting $w = \sqrt{\|y\|}$ and writing $a = u + iv$, the entire stability question becomes a planar phase estimate. The memory variable satisfies $u' = w$, and, with u as the independent variable, the unstable amplitude satisfies $dw/du = q/2$ for a coefficient $q \in [-1, 1]$. Once $u^2 + v^2 \geq w^2$, that region is invariant and w cannot increase. Soon afterward $dw/du \leq -1/4$, which bounds the total memory accumulation. This gives a uniform square-root stability modulus.

3 Counterexample

Theorem 1. *Let H_0 be any infinite-dimensional complex Hilbert space and put $H = H_0 \oplus \mathbb{C}$ with its Hilbert norm. There are a bounded generator $A \in \mathcal{B}(H)$, a continuous compact map $K : H \rightarrow H$, and a continuous map $G : H \rightarrow H$ such that:*

1. $\sigma_e(A) = \{1\}$, so the essential spectral bound of A is 1;
2. $K(0) = G(0) = 0$ and $\|G(x)\| \leq 2\|x\|^2$ for every $x \in H$;
3. equation (1) has a unique global classical solution for every initial state;
4. the zero solution is Lyapunov stable. More precisely, if $\eta = \|x(0)\| \leq 1$, then

$$\sup_{t \geq 0} \|x(t)\| \leq 50\sqrt{\eta}.$$

Proof. Write $x = (y, a) \in H_0 \oplus \mathbb{C}$ and $s = \|y\|$. Define

$$\begin{aligned} A(y, a) &= (y, 0), \\ K(y, a) &= (0, \sqrt{s}), \\ G(y, a) &= \left(-\frac{2|a|^2}{s + |a|^2} y, 0 \right), \end{aligned}$$

where the last expression is defined to be zero at $(0, 0)$. The operator A generates the semigroup $(y, a) \mapsto (e^t y, a)$. Moreover, $A - I$ has an infinite-dimensional kernel, while A is Fredholm: its kernel and cokernel are both one-dimensional. Since $A - \lambda I$ is invertible for $\lambda \notin \{0, 1\}$, it follows directly from the Fredholm definition that $\sigma_e(A) = \{1\}$.

The map K is continuous. On a bounded set its image is contained in a bounded subset of the one-dimensional space $\{0\} \oplus \mathbb{C}$, so it is precompact; hence K is a compact map. The apparent singularity in G is harmless, since

$$\|G(y, a)\| = \frac{2|a|^2 s}{s + |a|^2} \leq 2|a|^2 \leq 2\|(y, a)\|^2. \quad (2)$$

This estimate also proves continuity at the origin; continuity elsewhere is immediate (including points $(0, a)$, because the scalar coefficient is bounded by 2).

The differential equation is

$$y' = q(s, a)y, \quad a' = \sqrt{s}, \quad q(s, a) = \frac{s - |a|^2}{s + |a|^2}, \quad (3)$$

with the evident value at $(0, 0)$. Notice that $|q| \leq 1$. If $y(0) = 0$, Gronwall's inequality applied to the first equation forces $y \equiv 0$, and then a is constant. If $y(0) \neq 0$, the first equation gives $y(t) = \exp(\int_0^t q(\tau) d\tau)y(0)$, so y never vanishes and its direction is fixed. The variables $s = \|y\|$, $u = \operatorname{Re} a$, and $v = \operatorname{Im} a$ obey a smooth finite-dimensional system on $s > 0$:

$$s' = qs, \quad u' = \sqrt{s}, \quad v' = 0.$$

It is locally unique there. Since $s(0)e^{-t} \leq s(t) \leq s(0)e^t$ and $u' = \sqrt{s}$, no component can blow up in finite time. This proves unique global well-posedness.

It remains to prove stability. Assume $s(0) > 0$ and set

$$w = \sqrt{s}, \quad a = u + iv, \quad q = \frac{w^2 - u^2 - v^2}{w^2 + u^2 + v^2}.$$

Then

$$w' = \frac{q}{2}w, \quad u' = w, \quad v' = 0. \quad (4)$$

Thus u is strictly increasing and

$$\frac{dw}{du} = \frac{q}{2} \leq \frac{1}{2}. \quad (5)$$

First consider any part of the orbit with $u < 0$. Either u stays negative forever, in which case (5) already bounds w and all the state variables, or u first reaches zero with

$$w \leq w(0) + \frac{|u(0)|}{2}.$$

Accordingly, start (either at time zero if $u(0) \geq 0$, or at this first zero) with $u_* \geq 0$ and $w_* > 0$. Introduce the convenient initial bound

$$B = w(0) + |u(0)| + |v|;$$

then $u_* \leq B$ and $w_* \leq B$.

Let

$$D = u^2 + v^2 - w^2.$$

Where $D \geq 0$, one has $q \leq 0$ and

$$D' = 2uw - qw^2 = 2uw + \frac{Dw^2}{w^2 + u^2 + v^2} \geq 0.$$

Hence this region is forward invariant. Before the orbit enters it, (5) implies $w(u) \leq w_* + (u - u_*)/2$. By $u = u_* + 2w_*$ we have $u \geq 2w_*$ and $w \leq 2w_*$, so $D \geq 0$. Thus entry occurs at some (u_1, w_1) satisfying

$$u_1 \leq 3B, \quad w_1 \leq 2B. \quad (6)$$

After entry, w is non-increasing. If $u \geq \sqrt{3}w_1$, then $u^2 + v^2 \geq 3w^2$ and therefore $q \leq -1/2$. On that part of the orbit,

$$\frac{dw}{du} \leq -\frac{1}{4}.$$

Nonnegativity of w shows that, after u reaches $\max(u_1, \sqrt{3}w_1)$, it can increase by at most $4w_1$. Together with (6), this yields, for all $t \geq 0$,

$$w(t) \leq 2B, \quad |u(t)| \leq 12B, \quad |v(t)| \leq B. \quad (7)$$

Finally let $\eta = \|(y(0), a(0))\| \leq 1$. Since $s(0) \leq \eta$, $w(0) \leq \sqrt{\eta}$, and $|a(0)| \leq \eta$,

$$B \leq \sqrt{\eta} + \sqrt{2}\eta < \frac{5}{2}\sqrt{\eta}.$$

Equations (7) give $\|y(t)\| = w(t)^2 \leq 25\eta$ and $|a(t)| \leq 13B < \frac{65}{2}\sqrt{\eta}$. Consequently

$$\|(y(t), a(t))\| < 50\sqrt{\eta}$$

for all $t \geq 0$. The case $s(0) = 0$ was already seen to be stationary. Given $\varepsilon > 0$, choosing $\eta < \min\{1, (\varepsilon/50)^2\}$ proves Lyapunov stability of zero. $\square$

4 Verification and limitations

The proof is self-contained. The separate verifier report audits compactness, essential spectrum, well-posedness, and every phase-plane inequality. A small deterministic Runge–Kutta script in `code/simulate_reduced.py` checks representative initial phases and scales; it is a sanity check, not part of the proof.

The construction depends essentially on the sublinear memory law $\sqrt{\|y\|}$. The compact map K is continuous (indeed, $1/2$ -Hölder in the radial variable) but is not locally Lipschitz or Fréchet differentiable at zero. Therefore this packet does not answer a version imposing local Lipschitz, C^1 , or specified differentiability hypotheses on the compact perturbation. It also does not contradict the source’s positive instability theorems, which concern compact nonlinearities without the additional noncompact metrically small term G or invoke stronger local hypotheses.

5 Novelty check and review recommendation

A bounded search through 17 August 2026 covered the run’s four lightweight indexes, the source text and bibliography, the exact quoted sentence, the paper title and author names, and web/arXiv-oriented combinations of “continuous time,” “compact perturbation,” “metrically small,” “essential spectral bound,” and “stable zero solution.” It found the source paper and general linearized-instability literature, but no later paper giving this construction or explicitly resolving the quoted suspicion. The novelty claim is therefore limited to this explicit continuity-level counterexample.

Human review should focus on two points: whether “similar construction” is intended to include compact maps that are merely continuous at zero, as in the discrete setup; and the uniqueness argument at the non-Lipschitz equilibrium. The latter is supplied directly by the bounded scalar coefficient in the y equation, not by a general Banach-space existence theorem.

References

- [1] V. Müller, R. Schnaubelt, and Y. Tomilov, *On growth and instability for semilinear evolution equations: an abstract approach*, *Mathematische Annalen* 389 (2024), 3885–3933; arXiv:2211.15121v3.